

A PROJECT TITLE
ON
Renewable Energy Output Prediction Using XGBoost and LightGBM
project report submitted in partial fulfilment of the requirements for the Course
MACHINE LEARNING

BACHELOR OF TECHNOLOGY

in

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by

Name K. Rithwik Roll Number: 2520030428

Name K. Sai Satwik Roll Number: 2520030446

Under the Esteemed guidance of

Mr. Shaik Asif

Assistant Professor, CSE

Computer Science and Engineering



Koneru Lakshmaiah Education Foundation

(Deemed to be University estd. u/s. 3 of the UGC Act, 1956)

Off-Campus: Bachupally-Gandimaisamma Road, Bowrampet, Hyderabad, Telangana - 500 043.

Phone No: 7815926816, www.klh.edu.in



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DEPARTMENT OF Computer Science and Engineering

Name of Title: Renewable Energy Output Prediction Using XGBoost and LightGBM

1. Abstract:

Renewable energy has become one of the most important sources of electricity generation due to the increasing demand for clean, sustainable, and environmentally friendly energy. However, renewable energy sources such as solar and wind are highly dependent on weather conditions, making their power generation unpredictable. Accurate forecasting of renewable energy output is essential for efficient power grid management, energy distribution, and reducing operational costs. This project presents a machine learning-based approach for predicting renewable energy output using two advanced ensemble learning algorithms: XGBoost (Extreme Gradient Boosting) and LightGBM (Light Gradient Boosting Machine).

The project utilizes historical renewable energy generation data along with weather-related parameters such as temperature, humidity, wind speed, solar radiation, cloud cover, rainfall, and atmospheric pressure. The collected dataset is pre-processed by handling missing values, removing duplicates, and performing feature engineering to improve model performance. The processed data is then divided into training and testing sets for model development and evaluation.

Two regression models, XGBoost and LightGBM, are trained to learn the relationship between weather conditions and energy generation. Their performance is evaluated using regression metrics such as Mean Absolute Error (MAE), Mean Squared Error (MSE), Root Mean Squared Error (RMSE), and R^2 Score. The comparison helps identify the model that provides the most accurate predictions with better computational efficiency.

The developed model assists power generation companies, smart grid systems, and energy management organizations in improving operational planning, reducing energy wastage, and maintaining a reliable power supply. By comparing the performance of XGBoost and LightGBM using standard regression evaluation metrics, the project identifies the most suitable model for renewable energy forecasting. This work demonstrates how machine learning can contribute to the efficient management of renewable energy resources and support the transition toward a cleaner and more sustainable energy future.

2. INTRODUCTION:

The increasing global demand for electricity and the growing concern about climate change have accelerated the adoption of renewable energy sources such as solar and wind power. Unlike conventional energy sources, renewable energy generation depends heavily on environmental conditions, making its output highly variable and difficult to predict. This uncertainty creates challenges in maintaining a stable electricity supply, balancing power generation with demand, and ensuring efficient utilization of energy resources.

Machine Learning (ML) has emerged as an effective solution for forecasting renewable energy production by analysing historical weather and power generation data. Advanced ML algorithms can identify complex patterns and relationships between weather variables and energy output, leading to more accurate predictions than traditional forecasting methods.

This project focuses on predicting renewable energy output using two powerful gradient boosting algorithms: XGBoost and LightGBM. These algorithms are widely used because of their high prediction accuracy, ability to handle large datasets, and efficient training process. The system processes weather-related features such as temperature, humidity, wind speed, solar radiation, cloud cover, rainfall, and atmospheric pressure to estimate future energy generation.

3. SOFTWARE REQUIREMENTS:

Software	Purpose
Python 3.10 or above	Programming language used to develop the machine learning model
Jupyter Notebook / Google Collab	Writing, running, and testing the Python code
Pandas	Data loading, cleaning, and preprocessing
NumPy	Numerical computations and array operations
Scikit-learn	Data splitting, preprocessing, and model evaluation
XGBoost	Building the XGBoost prediction model
LightGBM	Building the LightGBM prediction model
Matplotlib	Visualizing data and model results

Signature of the Guide

Course Instructor